

Miniature Vibrating Fork Level Switch

For Liquid Application Model: MVFLS-800L



PRODUCT DESCRIPTION:

The MVFLS-800L Miniature Vibrating Fork Level Switch is a compact, reliable and cost-effective point level instrument specially designed for minimum and maximum level detection of liquids in tanks, vessels and process equipment. The compact vibrating fork is suitable for liquid media having a specific gravity from 0.785 to 1.5. Changes in the electrical properties of the process medium, such as conductivity and dielectric constant, have minimal effect on its operation, providing reliable and repeatable level detection. The MVFLS-800L can be used for a wide range of industrial liquids, including water-based liquids, chemicals, epoxy, resins, oils and petrochemicals, making it suitable for diverse process applications.

SALIENT FEATURES :



- Compact size with integral electronics for easy installation and space saving.
- Simple and easy calibration procedure for quick commissioning.
- No moving parts, eliminating mechanical wear and tear and ensuring long service life.
- 24 V DC power supply for convenient system integration.
- Standard PNP output available for direct connection to PLC and control systems.
- Short 68 mm fork design requiring minimum insertion length, suitable for small tanks, vessels and restricted installation spaces.
- Piezoelectric sensing technology for reliable and repeatable level detection.
- Suitable for minimum and maximum liquid level detection.
- Multiple process connection options: ½" BSP, ½" NPT, screwed and flanged mounting arrangements.
- LED status indication for clear identification of Normal and Alarm conditions.
- Compact vibrating fork construction suitable for a wide range of industrial liquid applications.
- Suitable for liquids with specific gravity from 0.785 to 1.5.
- Fast response and reliable switching for accurate point level detection.
- Suitable for water, chemicals, oils, epoxy, resins, petrochemicals and other industrial liquids.

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OPERATING PRINCIPLE:

The MVFLS-800L operates on the proven principle of vibrating fork technology. The tuning fork is continuously excited at its natural mechanical resonant frequency by integrated piezoelectric elements.

When the process liquid reaches and covers the fork tines, the liquid causes a change in the fork's vibration characteristics and damps the vibration amplitude. This change is detected by the electronic sensing circuit and converted into a reliable level switching signal.

The integrated electronics provides a PNP output corresponding to the detected level condition, which can be used for high/low level alarm, pump control, valve operation and other process control functions.

CONSTRUCTION SUITABLE FOR LIQUID APPLICATIONS:

The MVFLS-800L features a compact and robust tuning fork construction specifically designed for reliable point-level detection in liquid applications. The short 68 mm fork requires minimum insertion depth, making it suitable for small tanks, vessels and applications with limited installation space.

The fork is designed to provide stable and consistent vibration characteristics when immersed in the process liquid. Its compact construction allows easy installation through ½" BSP/NPT, screwed or flanged process connections.

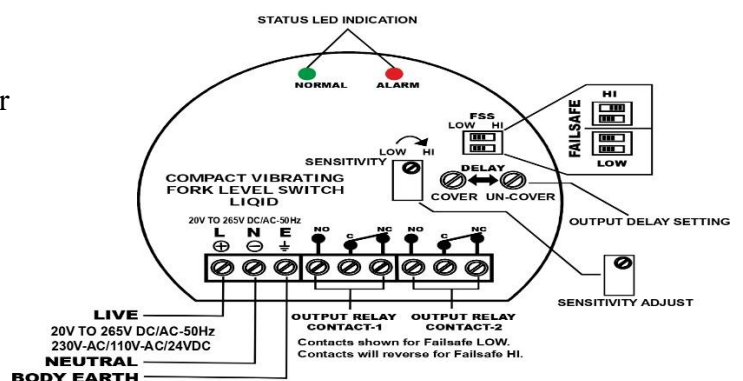
COMPONENTS OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE



- -----Cable Entry DIN Connector
- -----Acrylic
- -----SS Body
- -----Mounting Thread
- ----- Tuning Fork

RELAY OUTPUT UNIVERSAL /230 VAC/110VAC/24 VDC

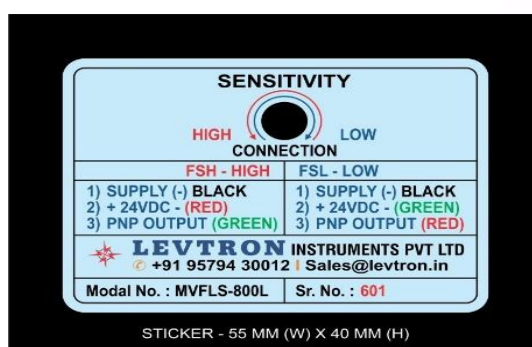
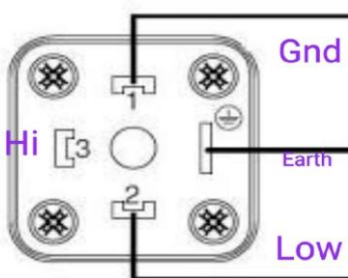
ELECTRICAL CONNECTION- UNIVERSAL



ELECTRICAL CONNECTION

POWER SUPPLY: 12 VDC TO 30 VDC

OUTPUT: PNP NO



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MVFLS-800L – Miniature Vibrating Fork Level Switch

Selection / Ordering Information

Selection	Code	Description
Model No.	MVFLS-800	Miniature Vibrating Fork Level Switch
Application	L	Liquid
Mounting Type	IN	Integral
Probe Type	C/W	Casting Fork / Wire Cut Fork
Probe Material	04 /4L/ 16/ 6L/ SM	SS304/SS304L / SS316/ SS316L / Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	AH/ WP / FL	*Acrylic /Weatherproof/ Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B0/B1 / B2 / B3/B4	½"/1" / 1½" / 2" BSP/ Special Mounting
	N0/N1 / N2 / N3/N4	½"/1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
MOC-Process Connection	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customized
Temperature	T1/TS	Max -20–150°C /Customer Specified Special Designed
High Temperature stand off	H8/HS	80MM/Special standoff Extension
Cable Gland	G0/G1/ G2 / G3/ G4	PG 9/ PG13.5 / ½"NPT / ¾" ET/
Ground / Reference	GS / GX	Standard / Special
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	68 mm TO Customized



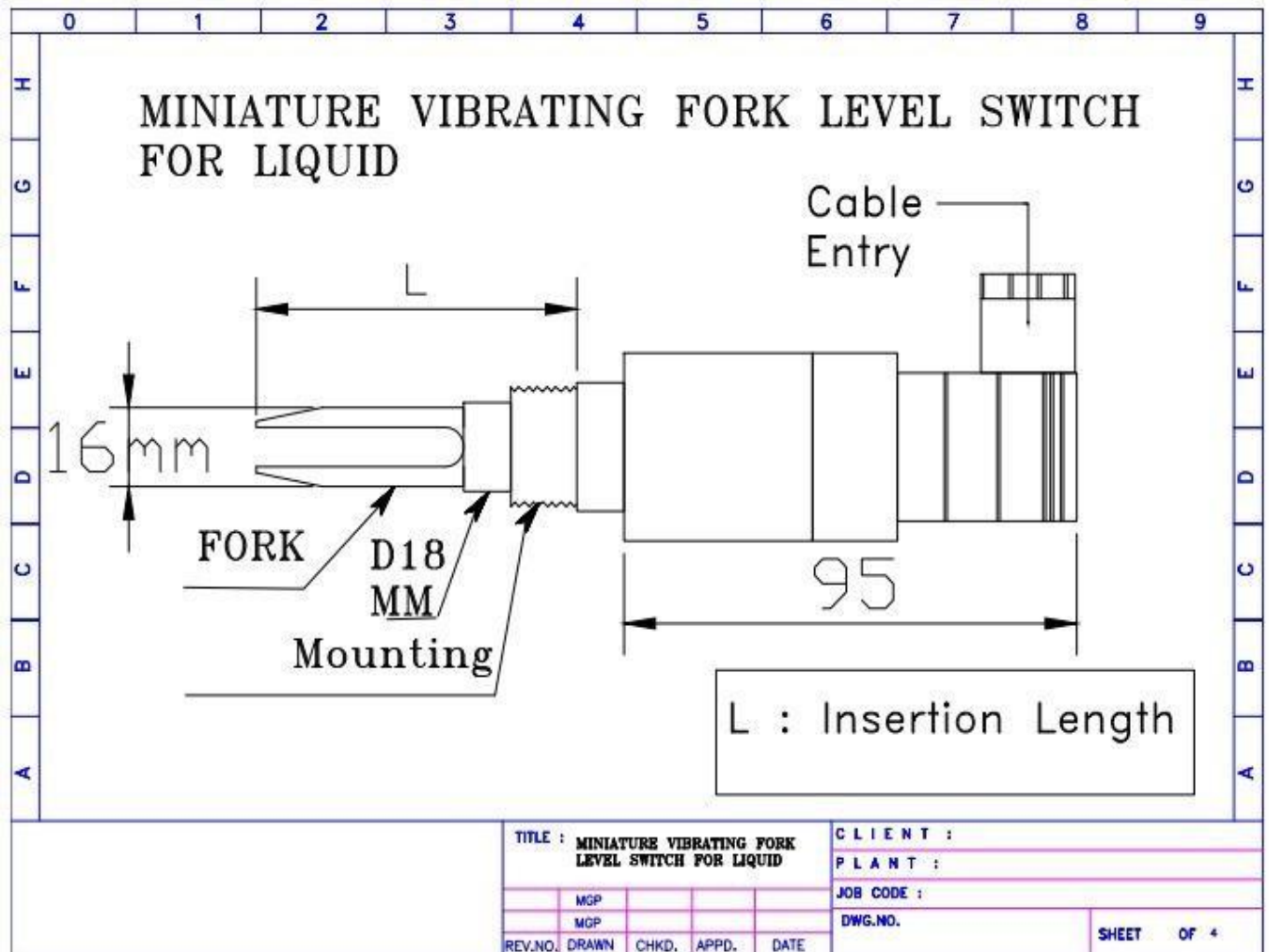
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LEVTRON
SENSING INSTRUMENTS



Mechanical Drawing:



Manufactured By:

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